

WALTER F. GEORGE PERTINENT DATA

GENERAL

Location (damsite), miles above mouth of Chattahoochee River (Spanning AL-GA border with features in Henry County, AL, and Clay County, GA)	75.0
Drainage area above damsite – square miles	7,460
Drainage area between damsite and West Point – square miles	4,020

RESERVOIR

Length – river miles	85.0
Top of conservation storage elevation summer – feet, NGVD29	190.0
Top of conservation storage elevation winter – feet, NGVD29	188.0
Peak pool for standard project flood – feet, NGVD29	200.1
Peak pool for spillway design flood – feet, NGVD29	206.6
Area at static full pool (elev. 190 feet NGVD29) – acres	41,800
Total volume at static full pool (elev. 190 feet NGVD29) – acre feet	884,572
Shore line length at static full pool (elev. 190 feet NGVD29) – miles	640

TAILWATER ELEVATIONS

Normal, one turbine operating (approx. 5,500 cfs)– feet, NGVD29	102.3 - 104.3
Normal, full powerhouse flow (approx. 26,000 cfs) – feet, NGVD29	109.3 - 126.1
Bankfull (approx. 65,000 cfs) – feet, NGVD29	130.2

DAM/EARTH DIKES

Total length – feet	12,128
Top elevation – feet, NGVD29	215.0

SPILLWAY SECTION

Total length including end piers – feet	708
Type of gates	Tainter
Size of gates, width x height – feet	42 x 29
Number of gates	14

LOCK

Maximum lift – feet	88.0
Chamber width – feet	82.0
Nominal chamber length – feet	450.0

POWER PLANT

Generating capacity (declared*) MW (4 units @ 42 MW)	168
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* Declared generating capacity is defined as the plant's operational capacity declared on a weekly basis to the power marketing agency. The value may vary slightly from week to week depending on factors such as head and cooling capabilities; values shown are the nominal values reported.

STREAM FLOW

Drainage area at dam site - square miles	7,460
Drainage area (dam to West Point Dam) - square miles	4,020
Minimum mean daily discharge, 19 October 1954, (affected by upstream regulation at existing power dams) - cfs	980
Minimum mean monthly flow (Oct. 1931) - cfs	1,758
Maximum mean monthly pre dam flow (Mar. 1929) - cfs	92,600
Maximum known pre dam flow (17 Mar. 1929) - cfs	203,000
Average annual pre dam flow (Aug. 1928 - Dec. 1961) - cfs	9,992
Average annual flow since dam completed (Apr. 1963 - Dec. 2013) - cfs	9,155
Discharge at bankfull stage (above George A. Andrews) - cfs	65,000
Discharge at bankfull stage (George A. Andrews to Jim Woodruff) - cfs	40,000
Minimum mean monthly flow at damsite post construction (Jul. 2008) - cfs	1,540
Maximum mean monthly flow at damsite post construction (Apr. 1964) - cfs	42,137
Maximum known flow at damsite post construction (17 Mar. 1990) - cfs	403,800

SPILLWAY DESIGN FLOOD

Total rainfall, Buford Dam to Walter F. George Dam - inches	16.10
Initial loss – inches	0.00
Average infiltration rate - inches per hour	0.05
Total storm runoff - inches	13.30
Total volume of storm runoff - acre feet	5,291,626
Peak rates of flow	
Natural flow at dam site - cfs	542,000
Inflow to full reservoir - cfs	767,800
Reservoir outflow - cfs	653,000
Base flow - cfs	13,000
Duration of flood - days	9

STANDARD PROJECT FLOOD

Total rainfall, Buford Dam to Walter F. George Dam - inches	8.37
Initial loss - inches	0.00
Average infiltration rate - inches per hour	0.05
Total storm runoff - inches	6.69
Total volume of storm runoff - acre feet	2,305,000
Peak rates of flow	
Natural flow at dam site - cfs	261,000
Inflow to full reservoir - cfs	380,300
Reservoir outflow - cfs	269,000
Base flow - cfs	13,000
Duration of flood - days	6

RESERVOIR

Pool elevations - feet NGVD29	
Maximum pool, spillway design flood (initial pool, elevation 190).	206.6
Maximum pool, standard project flood (initial pool elevation 190).	200.1
Top of Conservation Pool	
Summer	190.0
Winter	188.0
Minimum conservation pool	184.0
Storage volumes - acre feet	
Total volume at full summer pool (elevation 190 feet NGVD29)	884,572
Total volume at full winter pool (elevation 188 feet NGVD29)	802,172
Conservation storage, (190 to 184 feet NGVD29)	232,800
Total Inactive storage, (below elevation 184 feet NGVD29)	651,772
Reservoir areas - acres	
Full summer pool, (elevation 190 feet NGVD29)	41,800
Full winter pool, (elevation 188 feet NGVD29)	39,400
Bottom of conservation pool, (elevation 184 feet NGVD29)	34,600
Area within taking line - acres	
Purchased in fee simple	51,540
Right to inundate acquired by easement	27,809
Within Fort Benning military reservation	9,350
Riverbed	2,900
Total	91,599
Maximum elevation of clearing - feet NGVD29	192.0
Minimum elevation of clearing - feet NGVD29	175.0
Area cleared - acres	14,630
Length of shoreline at elevation 190 - miles	640
Length of reservoir at elevation 190 - miles	85

TAILWATER ELEVATIONS

Normal, no flow - feet NGVD29	101.2 - 102.1
Normal, one turbine operating (approx. 5,500 cfs) - feet NGVD29	102.3 - 104.3
Normal, full powerhouse flow (approx. 26,000 cfs)- feet NGVD29	109.3 - 126.1
Bankfull (approx. 65,000 cfs) - feet NGVD29	130.0
Highest known flood - feet NGVD29	158.0

DAM/EARTH DIKES

Total length - feet	12,128
Length on right bank - feet	6,124
Length on left bank - feet	6,004
Top elevation - feet NGVD29	215.0
Maximum height - feet	68.0
Top width - feet	30.0

SPILLWAY SECTION

Total length including end piers - feet	708
Net length - feet	588
Number of piers, including end piers	15
Width of piers - feet	8
Elevation of crest - feet NGVD29	163.0
Elevation of service bridge - feet NGVD29	208.08
Type of gates	Tainter
Size of gates - feet	42 X 29
Elevation top of gates - feet NGVD29	192.0
Number of gates	14
Elevation of high level apron - feet NGVD29	99.0
Number of bays, high level apron	8
Elevation of low level apron - feet NGVD29	94.0
Number of bays, low level apron	6
Length of high level apron - feet	89.5
Length of low level apron - feet	86.0
Maximum allowable head differential - feet	88.0

LOCK

Maximum lift - feet	88.0
Chamber width - feet	82.0
Nominal chamber length - feet	450
Distance center to center of pintles - feet	505
Freeboard at lower guide and guard was when lock becomes inoperative - feet	1
Percent of time lock will be inoperative	58
Type of gates	Miter
Height of upper gate - feet	36.0
Height of lower gate - feet	108.0
Type of culvert valves	Reverse Tainter
Dimensions of culverts - feet	12 X 12
Elevations - feet NGVD29	
Top of upper guide and guard walls	197.0
Top of walls at upper gate blocks	208.0
Top of chamber walls	197.0
Top of lower guide and guard walls	147.0
Top of upper miter sill	172.0
Top of lower miter sill	89.0
Chamber floor	88.0
Elevation top of stoplogs, downstream	89.0
Elevation top of stoplogs, upstream	172.0
Total time required to fill lock chamber, starting with valves closed - minutes	13.0
Total time required to empty lock chamber, starting with valves closed - minutes	12.5

POWER PLANT

Length of foundation - feet	335
Width of foundation of powerhouse and intake - feet	168
Number of units	4
Spacing of units - feet	67
Type of turbines	Adjustable blade, propeller/type
Type of scroll case	Concrete
Elevation of distributor - feet NGVD29	103.0
Elevation of bottom of draft tube - feet NGVD29	57.5
Length of draft tube - feet	66.5
Type of draft-tube gate	Sliding
Number of draft-tube gates per unit	2
Size of draft-tube gates - feet	24 x 22
Type of operation for draft-tube gates	Gantry Crane
Generator rating, each unit - kW	42,000
Generator rating, each unit - kVA	46,666
Generator speed - rpm	112.5
Generator electrical characteristics	3 phase, 60 cycle 13,800 volts
Location of switchyard	Right bank, downstream
Elevation of switchyard - feet NGVD29	160.5
Transmission voltage - kV	115

POWER DATA

Rated net head - feet	88.0
Rated net head - test	70.0
Minimum net head - feet	37.0
Plant output at rated net head	
Installed capacity at rated power factor - MW	168
Installed capacity at unity power factor - MW	184
Average annual energy (1964 – 2013) - MWH	428,881
Tailwater elevations - feet NGVD29	
All turbines 168 MW (approx. 26,000 cfs)	109.3 - 126.1
Three turbines 126 MW (approx. 19,500 cfs)	107.1 - 110.8
Two turbines 84 MW (approx. 12,000 cfs)	105.4 - 107.8
One turbine 42 MW (approx. 5,500 cfs)	102.3 - 104.3
No flow	101.2 - 102.1